

Healthcare AI Proof-of-Concept

A simulated digital representation of health state, trends, anomalies and What-If scenarios.

From fragmented metrics to an understandable health state

Health measurements can be difficult to interpret when viewed individually. A digital twin provides a consolidated representation

A Digital Twin for Health Monitoring

- Current health-state dashboard
- Historical trend visualization
- Explainable anomaly detection
- Health insights
- What-If lifestyle simulation
- REST API architecture

Digital Twin Data Flow

React + TypeScript Frontend



REST API



FastAPI Backend



Analytics Simulation



Synthetic / Demo Health Data

What the prototype demonstrates

- 01 Digital Twin State
- 02 Health Metrics
- 03 Historical Trends
- 04 Anomaly Detection
- 05 Explainable Insights
- 06 What-If Simulation

Built as a full-stack AI prototype

Frontend: React, TypeScript, Vite, Tailwind CSS

Backend: Python, FastAPI, Uvicorn

Analytics: Pandas, NumPy, Scikit-learn

Communication: REST APIs

Data: Synthetic / demonstration data

Five-minute demonstration flow

Dashboard → Current Twin State → Historical Trends → Anomalies → Explainable Insights → What-If Simulation → Architect

Safety, privacy and transparency

- Synthetic/demo data
- Explainable outputs
- No medical diagnosis claims
- No treatment recommendations
- Production use requires validation, security, consent and governance

From proof-of-concept to validated platform

Wearable integration

Longitudinal patient modeling

Validated ML forecasting

Privacy-preserving learning

Clinician dashboards

Secure authentication and audit logging

Clinical and regulatory validation

Digital Twin Health

A proof-of-concept demonstrating how health data, analytics and simulation can work together to create an understandable digital twin

GitHub: github.com/Maddy67bh/digital-twin-health

Thank you